

Causal inference with conjoint

Sliders and switches

**Categorical
variable**



**Continuous
variable**



**Many
simultaneous
continuous
variables**



**Many
simultaneous
categorical
variables**


```

1 library(tidyverse)
2 library(marginaleffects)
3 library(parameters)
4
5 penguins <- penguins |> drop_na(sex)
6
7 model1 <- lm(body_mass ~ flipper_len, data = penguins)
8 model2 <- lm(body_mass ~ flipper_len + species, data = penguins)

```

```
1 model_parameters(model1)
```

Parameter	Coefficient	SE	95% CI	t(331)	p
(Intercept)	-5872.09	310.29	[-6482.47, -5261.71]	-18.92	< .001
flipper len	50.15	1.54	[47.12, 53.18]	32.56	< .001

```
1 model_parameters(model2)
```

Parameter	Coefficient	SE	95% CI	t(329)	p
(Intercept)	-4013.18	586.25	[-5166.44, -2859.92]	-6.85	< .001
flipper len	40.61	3.08	[34.55, 46.66]	13.19	< .001
species [Chinstrap]	-205.38	57.57	[-318.62, -92.13]	-3.57	< .001
species [Gentoo]	284.52	95.43	[96.79, 472.25]	2.98	0.003

{marginaleffects}

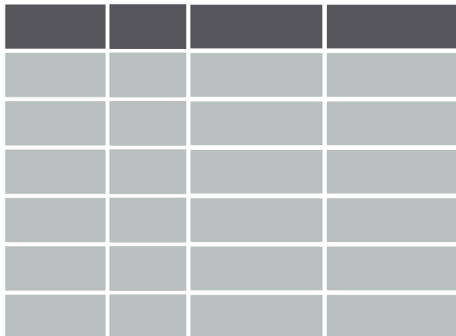
Quit working with raw coefficients!

Arel-Bundock, Vincent, Noah Greifer, and Andrew Heiss. 2024. “How to Interpret Statistical Models Using marginaleffects for R and Python.” Journal of Statistical Software 111 (9): 1–32. <https://doi.org/10.18637/jss.v111.i09>.

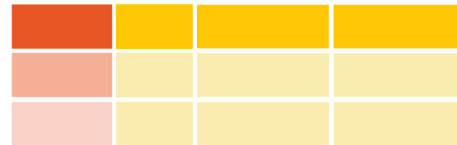
Arel-Bundock, Vincent. 2026. Model to Meaning: How to Interpret Statistical Models with R and Python. A Chapman & Hall Book. CRC Press, Taylor & Francis Group. <https://doi.org/10.1201/9781003560333>, <https://marginaleffects.com/>

Marginal effects at user-specified or representative values (MER)

Data



Reference grid

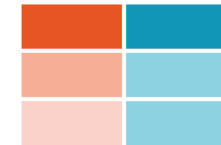


One or more covariates are set to **specific values** (i.e. $x_1 = c(10, 50, 100)$) while all others are set to **typical or average values**

Model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots$$

Fitted results



Marginal details for **each row** of the reference grid come from the **fitted results**

1. **Quantity:** What is the quantity of interest? Do we want to report a prediction or a function of predictions (average, difference, ratio, derivative, etc.)?
2. **Grid:** What predictor values are we interested in? Do we want to report estimates for the units in our dataset, or for hypothetical or representative individuals?
3. **Aggregation:** Do we report estimates for every observation in the grid or a global summary?
4. **Uncertainty:** How do we quantify uncertainty about our estimates?
5. **Test:** Which (non-)linear hypothesis or equivalence tests do we conduct?

```
1 model_parameters(model1)
```

Parameter	Coefficient	SE	95% CI	t(331)	p
(Intercept)	-5872.09	310.29	[-6482.47, -5261.71]	-18.92	< .001
flipper len	50.15	1.54	[47.12, 53.18]	32.56	< .001

```
1 avg_comparisons(model1)
```

Estimate	Std. Error	z	Pr(> z)	S	2.5 %	97.5 %
50.2	1.54	32.6	<0.001	770.2	47.1	53.2

Term: flipper_len

Type: response

Comparison: +1


```
1 model_parameters(model2)
```

Parameter	Coefficient	SE	95% CI	t(329)	p
(Intercept)	-4013.18	586.25	[-5166.44, -2859.92]	-6.85	< .001
flipper len	40.61	3.08	[34.55, 46.66]	13.19	< .001
species [Chinstrap]	-205.38	57.57	[-318.62, -92.13]	-3.57	< .001
species [Gentoo]	284.52	95.43	[96.79, 472.25]	2.98	0.003

```
1 avg_comparisons(model2)
```

Term	Contrast	Estimate	Std. Error	z	Pr(> z)	S	2.5 %	97.5 %
flipper_len +1		40.6	3.08	13.19	< 0.001	129.5	34.6	46.6
species	Chinstrap - Adelie	-205.4	57.57	-3.57	< 0.001	11.4	-318.2	-92.5
species	Gentoo - Adelie	284.5	95.43	2.98	0.00287	8.4	97.5	471.6

Type: response

```
1 model3 <- lm(body_mass ~ flipper_len * species, data = penguins)
2 model_parameters(model3)
```

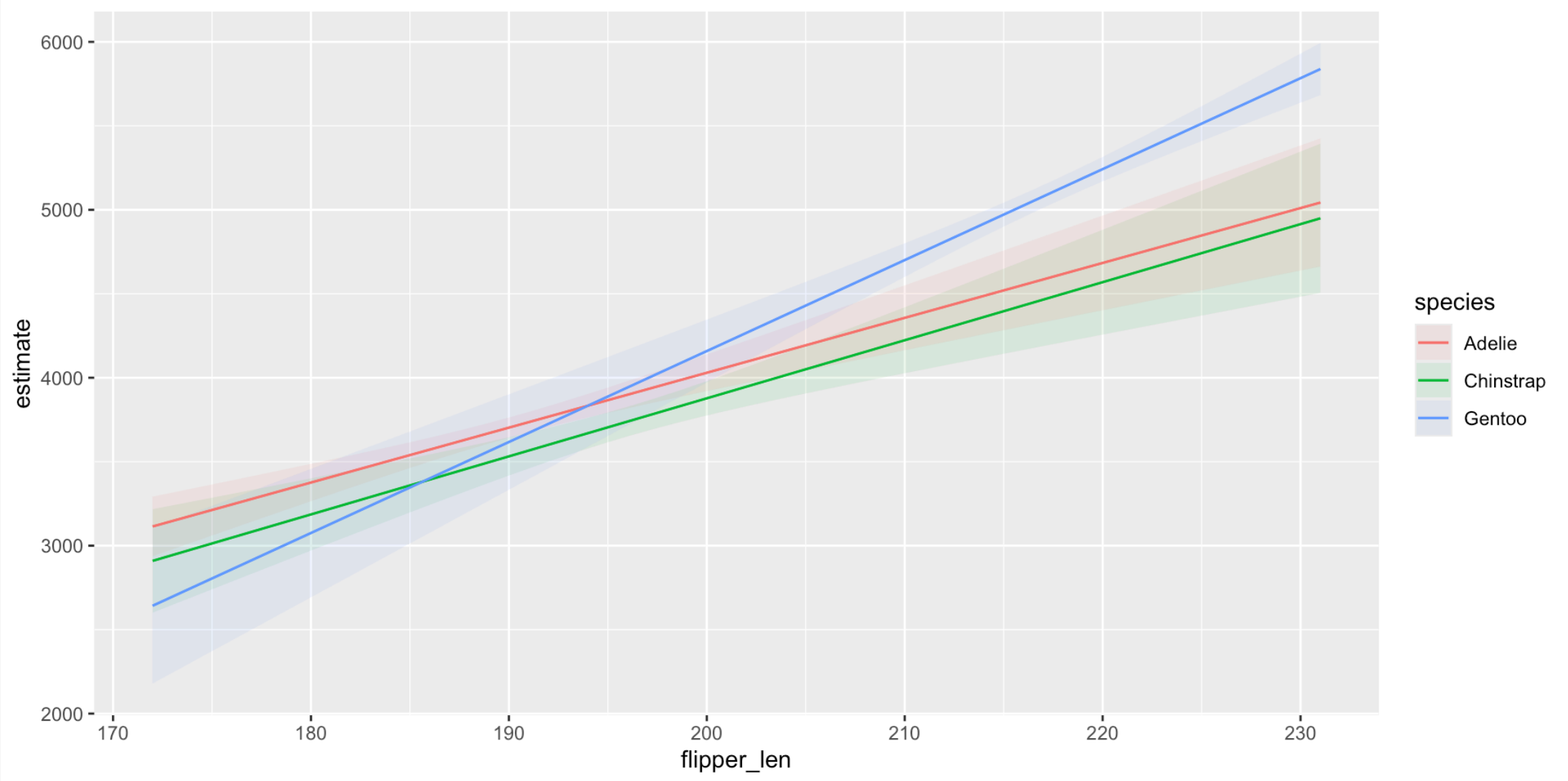
Parameter	Coefficient	SE	95% CI	t(327)	p

--					
(Intercept)	-2508.09	892.41	[-4263.68, -752.49]	-2.81	0.005
flipper len	32.69	4.69	[23.46, 41.92]	6.97	<.001
species [Chinstrap]	-529.11	1525.11	[-3529.37, 2471.16]	-0.35	0.729
species [Gentoo]	-4166.12	1431.58	[-6982.39, -1349.84]	-2.91	<.001

```
1 avg_comparisons(model3, by = "species")
```

Term	Contrast	species	Estimate	Std. Error	z	Pr(> z)	S	2.5 %	97.5 %
flipper_len +1		Adelie	32.7	4.69	6.967	< 0.001	38.2	23.5	41.9
flipper_len +1		Chinstrap	34.6	6.31	5.478	< 0.001	24.5	22.2	46.9
flipper_len +1		Gentoo	54.2	5.15	10.516	< 0.001	83.5	44.1	64.3
species	Chinstrap - Adelie	Adelie	-170.9	65.04	-2.627	0.00861	6.9	-298.3	-43.4
species	Gentoo - Adelie	Adelie	-83.4	146.97	-0.567	0.57052	0.8	-371.4	204.7
species	Chinstrap - Adelie	Chinstrap	-160.1	60.39	-2.651	0.00803	7.0	-278.4	-41.7
species	Gentoo - Adelie	Chinstrap	39.5	122.28	0.323	0.74675	0.4	-200.2	279.2
species	Chinstrap - Adelie	Gentoo	-119.7	193.37	-0.619	0.53580	0.9	-498.7	259.3
species	Gentoo - Adelie	Gentoo	499.3	135.18	3.694	< 0.001	12.1	234.4	764.3

```
1 plot_predictions(model3, condition = c("flipper_len", "species"))
```



What are marginal means?

OLS with categorical predictors

```
1 model <- lm(body_mass ~ species + sex, data = penguins)
```

Predictions across balanced grid

```
1 grid <- datagrid(model = model, species = unique, sex = unique)
2
3 preds <- predictions(model, grid)
```

```
1 preds |>
2   as_tibble() |>
3   select(species, sex, estimate)
```

```
# A tibble: 6 × 3
  species    sex estimate
  <fct>    <fct>    <dbl>
1 Adelie  female    3372.
2 Adelie  male      4040.
3 Chinstrap female    3399.
4 Chinstrap male      4067.
5 Gentoo  female    4750.
6 Gentoo  male      5418.
```

Species	Sex	
	Female	Male
Adelie	3372	4040
Chinstrap	3399	4067
Gentoo	4750	5418

Means in the margins

Species	Sex		Marginal mean
	Female	Male	
Adelie	3372	4040	3706 (3372 + 4040) / 2
Chinstrap	3399	4067	3733 (3399 + 4067) / 2
Gentoo	4750	5418	5084 (4750 + 5418) / 2
Marginal mean	3841 (3372 + 3399 + 4750) / 3	4508 (4040 + 4067 + 5418) / 3	

For the sex-specific averages, all the between-species variation is “marginalized out” or accounted for; for the species-specific averages, all the between-sex variation is similarly marginalized out.

Differences in marginal means

Sex				
Species	Female	Male	Marginal mean	Δ
Adelie	3372	4040	3706 (3372 + 4040) / 2	—
Chinstrap	3399	4067	3733 (3399 + 4067) / 2	27 Chinstrap–Adelie
Gentoo	4750	5418	5084 (4750 + 5418) / 2	1378 Gentoo–Adelie
Marginal mean	3841 (3372 + 3399 + 4750) / 3	4508 (4040 + 4067 + 5418) / 3		
Δ	—	668 Male–Female		

Coefficients and predictions

```
1 parameters::model_parameters(model)
```

Parameter	Coefficient	SE	95% CI	t(329)	p
(Intercept)	3372.39	31.43	[3310.56, 3434.21]	107.31	< .001
species [Chinstrap]	26.92	46.48	[-64.52, 118.37]	0.58	0.563
species [Gentoo]	1377.86	39.10	[1300.93, 1454.78]	35.24	< .001
sex [male]	667.56	34.70	[599.29, 735.82]	19.24	< .001

```
1 avg_predictions(model, by = "species", newdata = "balanced")
```

species	Estimate	Std. Error	z	Pr(> z)	S	2.5 %	97.5 %
Adelie	3706	26.2	141.4	<0.001	Inf	3655	3758
Chinstrap	3733	38.4	97.2	<0.001	Inf	3658	3808
Gentoo	5084	29.0	175.1	<0.001	Inf	5027	5141

Type: response

```
1 avg_predictions(model, by = "sex", newdata = "balanced")
```

sex	Estimate	Std. Error	z	Pr(> z)	S	2.5 %	97.5 %
female	3841	25.3	152	<0.001	Inf	3791	3890
male	4508	25.1	180	<0.001	Inf	4459	4557

Type: response